

[Problems](#)[Status](#)[Standings](#)

Prime Ring Problem

Tags:

Time Limit: 10 s Memory Limit: 32 MB
Submission: 137 AC: 30 Score: 98.69

Submit

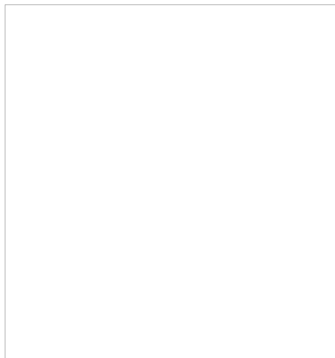
Codes

AI Code Tips

Description

A ring is composed of n circles as shown in diagram. Put natural number 1, 2, ..., n into each circle separately, and the sum of numbers in two adjacent circles should be a prime.

Note: the number of first circle should always be 1.



Input

n ($0 < n < 20$)

Output

The output format is shown as sample below. Each row represents a series of circle numbers in the ring beginning from 1 clockwise and anticlockwise. The order of numbers must satisfy the above requirements. Print solutions in lexicographical order.

You are to write a program that completes above process.

Print a blank line after each case.

Samples

input

Copy

6
8

output

Copy

Case 1:

1 4 3 2 5 6

1 6 5 2 3 4

Case 2:

1 2 3 8 5 6 7 4

1 2 5 8 3 4 7 6

1 4 7 6 5 8 3 2

1 6 7 4 3 8 5 2

Submit

Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

[--- Maintainer List ---](#)

Server Time: 2025-8-26 15:11:32